$$\Rightarrow \begin{bmatrix} 8 & 6 \\ 5 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \quad (\text{Defn. of } Ax=b)$$

By Theorem 5, $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 & -3 \\ -5/2 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 2\cdot 2+(-3)\cdot(-1) \\ (-5/2)\cdot 2+4\cdot(-1) \end{bmatrix}$ (Row-Vector Rule)

$$= \begin{bmatrix} 7 \\ -9 \end{bmatrix} \quad \boxed{\therefore x_1 = 7, x_2 = -9}$$

7. Let $A = \begin{bmatrix} 1 & 2 \\ 5 & 12 \end{bmatrix}$, $b_1 = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$, $b_2 = \begin{bmatrix} 1 \\ -5 \end{bmatrix}$, $b_3 = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$, and $b_4 = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$.

a) Find A^{-1}, and use it to solve the four equations $Ax=b_1$, $Ax=b_2$, $Ax=b_3$, $Ax=b_4$.

b) The four equations in part (a) can be solved by the same set of row operations, since the coefficient matrix is the same in each case. Solve the four equations in part (a) by row reducing the augmented matrix $[A\ b_1\ b_2\ b_3\ b_4]$

2 $\boxed{\text{Solution:}}$ (a). Determinant: $ad-bc = 1(12)-2(5) = 2$

By Theorem 4, $A^{-1} = \frac{1}{2}\begin{bmatrix} 12 & -2 \\ -5 & 1 \end{bmatrix} = \begin{bmatrix} 6 & -1 \\ -5/2 & 1/2 \end{bmatrix}$

We shall apply Theorem 5 and Row-Vector Rule for Computing Ax for solving the equations $Ax=b_1$, $Ax=b_2$, $Ax=b_3$, $Ax=b_4$.

$$A^{-1}b_1 = \begin{bmatrix} 6 & -1 \\ -5/2 & 1/2 \end{bmatrix} \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 6\cdot(-1)+(-1)\cdot 3 \\ (-5/2)\cdot(-1)+(1/2)\cdot 3 \end{bmatrix} = \begin{bmatrix} -9 \\ 4 \end{bmatrix}$$

$$A^{-1}b_2 = \begin{bmatrix} 6 & -1 \\ -5/2 & 1/2 \end{bmatrix} \begin{bmatrix} 1 \\ -5 \end{bmatrix} = \begin{bmatrix} 6\cdot 1+(-1)\cdot(-5) \\ (-5/2)\cdot 1+1/2\cdot(-5) \end{bmatrix} = \begin{bmatrix} 11 \\ -5 \end{bmatrix}$$

$$A^{-1}b_3 = \begin{bmatrix} 6 & -1 \\ -5/2 & 1/2 \end{bmatrix} \begin{bmatrix} 2 \\ 6 \end{bmatrix} = \begin{bmatrix} 6\cdot 2+(-1)\cdot 6 \\ (-5/2)\cdot 2+(1/2)\cdot 6 \end{bmatrix} = \begin{bmatrix} 6 \\ -2 \end{bmatrix}$$

$$A^{-1}b_4 = \begin{bmatrix} 6 & -1 \\ -5/2 & 1/2 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 6\cdot 3+(-1)\cdot 5 \\ (-5/2)\cdot 3+(1/2)\cdot 5 \end{bmatrix} = \begin{bmatrix} 13 \\ -5 \end{bmatrix}$$